

RESEARCH ARTICLE

Post-treatment of pretreated paper mill effluent via the electro-persulfate process: Optimization with response surface methodology approach

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ABSTRACT

This study examined the post-treatment of pre-treated paper industry wastewater obtained via electrocoagulation (EC) using Fe-Fe electrodes. The pre-treated wastewater was further treated with peroxymonosulfate (PMS) activation using platinum (Pt)/titanium dioxide (TiO₂) anode and graphite cathode. Process optimization was conducted using a response surface methodology (RSM) based on a Box-Behnken design (BBD) to systematically evaluate the effects of applied current, initial pH, PMS dosage, and reaction time on COD removal efficiency and energy consumption. The ANOVA results confirmed the statistical significance and strong predictive ability of the developed model. The model exhibited high R² values (R² = 0.9615 and Adj R² = 0.9230), an adequate signal-to-noise ratio, and a non-significant lack of fit. The optimization results of the electro-persulfate (EP) process show that the main parameters influencing COD removal and specific energy consumption (SEC) are the applied current and time. These results indicate that the model reliably optimizes COD removal. Optimal conditions for maximizing COD removal were predicted to be a current of 4.32 A, a pH of 8, a PMS dose of 16.3 mM, and a reaction time of 80 min. These conditions were expected to achieve 99.9% COD removal. However, the experimental results showed 92.3% ± 2.3% COD removal, 86.3% ± 2.8% tannin/lignin removal, and 57.2% ± 6.03% additional color removal. These results underscore the EP process's effectiveness and flexibility for industrial wastewater post-treatment, allowing for adjustments in treatment performance and energy requirements.

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INTRODUCTION

The paper industry produces cardboard, paper, and other cellulose-based products from lignocellulosic materials, such as wood and recycled paper. Paper mills are a significant and important global industry with high production and demand potential. Therefore, it is a crucial sector to consider in terms of environmental impact, particularly with regard to the consumption of natural resources such as wood and water, energy usage, and wastewater production. A significant amount of wastewater is produced in this industry during the production process, typically ranging from 75 to 275 m³ per ton of product. Depending on the raw material, product type, and applied process, this volume includes 20-25 m³ from the pulping process and 80-100 m³ from the bleaching process. While these values are subject to variation, it is important to note that they are well-established and widely accepted within the industry [1, 2].

Wastewater generated during post-production contains high levels of organic matter, suspended solids, and color.

Wastewater leaving the production line contains highly toxic and recalcitrant compounds, such as phenols, chlorinated lignosulfonic acids, tannins, lignins, fatty acids, resin acids, absorbable organic halides, xenobiotics, hydrocarbons, dibenzodioxins, and dibenzofurans. According to several sources, polymerized tannins, lignins, and their derivatives are not biodegradable and are the main cause of color in wastewater [3-6]. Additionally, tannins have been found to absorb heat and light, which can lead to a decrease in dissolved oxygen (DO) and have a negative impact on the flora and fauna in the aquatic environment [7]. Furthermore, it has been observed that some substances, such as resin acids, long chain fatty acids and organochlorides, can be harmful to aquatic organisms and accumulate in the aquatic food chain [7]. Hence, it is imperative to process wastewater to eliminate these persistent pollutants and avert their detrimental impact on the environment prior to releasing them into the receiving media. Proper treatment of these pollutants is necessary before discharge to prevent their

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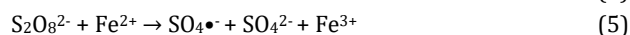
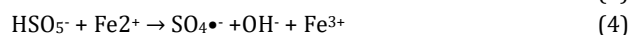
negative impact on the biodegradability of wastewater and the aquatic environment [8, 9].

Conventional methods used for the treatment of industrial wastewater from paper mills, such as primary treatment, leaching and aerobic biological treatment [10], are not sufficient to remove recalcitrant substances and impurities related to decomposition-resistant lignin [11]. Common physical treatment methods (eg, adsorption, filtration, leaching, sedimentation) remove only a small portion of the organic material and do not completely degrade it [12]. Conventional treatment approaches are therefore supported by advanced oxidation processes (AOPs) to meet emission standards and improve water quality and biodegradability. Electrochemical methods are preferable for removing suspended solids and colloidal substances, as classical methods are inadequate. This is a crucial step towards a more sustainable and environmentally friendly paper mill industry.

Electrochemical treatment technologies have been extensively researched as a viable alternative and have been proven to be highly effective in treating wastewaters containing recalcitrant and resistant organic pollutants [13-15]. In recent years, iron (Fe) has emerged as one of the most suitable substances for application in wastewater treatment due to its abundance and versatile role as both a catalyst and coagulant in electrochemical processes. Electrochemical processes based on Fe are classified as electrocoagulation (EC), electro-Fenton (EF) or a combination of both [16, 17]. Electro-oxidation (EO) is an advanced electrochemical treatment process in which organic pollutants degrade through direct electron transfer at the anode surface or indirectly via in situ generated oxidizing species [18]. This process primarily relies on the formation of hydroxyl radicals ($\bullet\text{OH}$) on the surface of insoluble anodes with a high oxygen overpotential. These highly reactive radicals can promote the partial degradation or complete mineralization of organic contaminants in wastewater [19]. In recent years, sulfate radical-based electro-oxidation, generally called electro-persulfate process, has gained attention as a method of driving pollutant degradation. In this process, sulfate radicals ($\text{SO}_4\bullet^-$) are generated, either independently or in combination with hydroxyl radicals [20].

Electro-persulfate processes involve the electrochemical activation of persulfate species, such as peroxymonosulfate (PMS, HSO_5^- , $E^\circ = 1.82 \text{ V}$) and peroxydisulfate (PDS, $\text{S}_2\text{O}_8^{2-}$, $E^\circ = 2.01 \text{ V}$), to produce highly reactive sulfate radicals ($\text{SO}_4\bullet^-$) [21]. Both PMS and PDS contain an O-O peroxide bond that can be cleaved through electron transfer, resulting in the formation of sulfate radicals (Eqs. (1) and (2)) [22, 23]. Persulfate (PS) can be added to the system externally or generated in situ via electrochemical reactions. Whether produced electrochemically or added externally, PS generates sulfate radicals ($\text{SO}_4\bullet^-$) upon activation. These radicals are highly oxidizing and possess a higher redox potential than hydroxyl radicals. This enables the non-selective and efficient degradation of various organic compounds. The longer half-life of $\text{SO}_4\bullet^-$ compared to $\bullet\text{OH}$ improves mass transfer and contact with pollutants. Secondary radicals formed from $\text{SO}_4\bullet^-$ reactions with water, such as $\bullet\text{OH}$, further enhance the overall oxidation efficiency [21, 22]. In this process, a persulfate can be activated using a metal electrode, such as iron, silver, zinc, copper, or manganese, to form a sulfate radical. However, iron is the most preferred anode material due to its lower cost and relative nontoxicity. When sacrificial iron anodes are used, iron (II) ions (Fe^{2+}) are generated via anodic dissolution, as described by Eqs. (3-5) [20]. Due to their high oxidation potential and chemical stability, platinum

(Pt) anodes have been used in electro-persulfate processes to treat various industrial wastewater, washing machine effluent dyeing [24], military ammunition plants, and washing machines [25], as well as other polluted waters [26].



Despite extensive studies on electrochemical and electro-oxidation processes for industrial wastewater treatment, limited research has focused on the integration of iron-based electrocoagulation as a pretreatment step followed by electro-persulfate activation using TiO_2/Pt anodes for paper mill wastewater. To the best of our knowledge, no previous study has systematically investigated the integration of Fe-Fe electrocoagulation pretreatment with PMS-activated electro-oxidation using TiO_2/Pt anodes for the treatment of paper industry wastewater. Therefore, the main objective of this study is to investigate the efficiency of an integrated electrochemical treatment process for the removal of recalcitrant organic pollutants from paper industry wastewater. The proposed treatment configuration aims to enhance COD removal efficiency, offering a technically effective and scalable alternative for the treatment of recalcitrant paper industry effluents.

MATERIALS AND METHODS

Paper Mill Wastewater Characterization

In this study, paper mill effluent was collected from a waste paper board manufacturing plant in Istanbul. The raw effluent was initially treated by EC process (achieving 35.7% COD removal efficiency with Fe-Fe electrodes under 2 A applied current, pH 6, 70 min, 0.12 M NaCl) and as it did not meet the discharge criteria to the receiving environment and sewerage, a post-treatment study was carried out using PMS supported EO process. Table 1 shows the characteristics of the EC pre-treated effluent.

All analyses except for tannin/lignin were conducted in accordance with Standard Methods [27]. Colour and tannin/lignin concentrations were measured using a Hach Lange DR 5000 spectrophotometer, colour at a wavelength of 550 nm and tannin/lignin after the addition of Hach TanniVer 3 reagent. All chemicals used throughout the study were of analytical grade.

Table 1. EC-pretreated paper mill wastewater characterization

Parameter	Average value $\pm$ Standard deviation
pH	9.82 $\pm$ 0.2
Conductivity (mS/cm)	17.71 $\pm$ 0.2
COD (mg/L)	3002 $\pm$ 55
Tannin/Lignin (mg/L)	4.75 $\pm$ 1.3
Color (Pt-Co)	86.5 $\pm$ 3

Experimental Set-up

The pre-treated wastewater was further treated by EP process using a TiO_2/Pt anode and a graphite cathode. Graphite was chosen as the cathode material because of its high electrical conductivity, chemical stability, and inert behavior under electrochemical conditions. Unlike metal cathodes, graphite does not dissolve or become passivated, which ensures stable electrochemical performance and lower energy consumption [28]. Graphite cathodes are widely used in electrochemical oxidation and electro-persulfate systems because they minimize parasitic reactions and electrode fouling [29]. Batch experiments were carried out in a Plexiglas reactor ($75 \times 75 \times 35$ mm) with a working volume of 100 mL. Continuous mixing was maintained using a magnetic stirrer. A digital DC power supply (ADC-3306D) was used to apply the current and the electrodes were placed 2 cm apart inside the reactor. The setup of the EP reactor is shown in Figure 1. During the experiments, samples were taken periodically from the reactor to evaluate the COD removal performance. The study investigated the influence of several operating parameters, including applied current (1-5 A), initial pH (2-10), PMS dosage (5-25 mM) and reaction time (10-90 min). The pH range of 2–6–10 was chosen to evaluate the effects of acidic, neutral, and alkaline conditions on persulfate activation and oxidation performance. Acidic conditions ($\text{pH} \approx 2$) favor the stability and reactivity of sulfate radicals ($\text{SO}_4^{\bullet-}$). Near-neutral conditions ($\text{pH} \approx 6$), which represent the characteristics of typical paper mill wastewater and practical operating conditions, were also included. Alkaline conditions ($\text{pH} \approx 10$) were included to evaluate the shift from sulfate radical-dominated oxidation to hydroxyl radical-dominated pathways resulting from radical conversion reactions. Thus, this pH range allows for a comprehensive evaluation of oxidation kinetics, dominant radical species, and process efficiency under different operational environments.

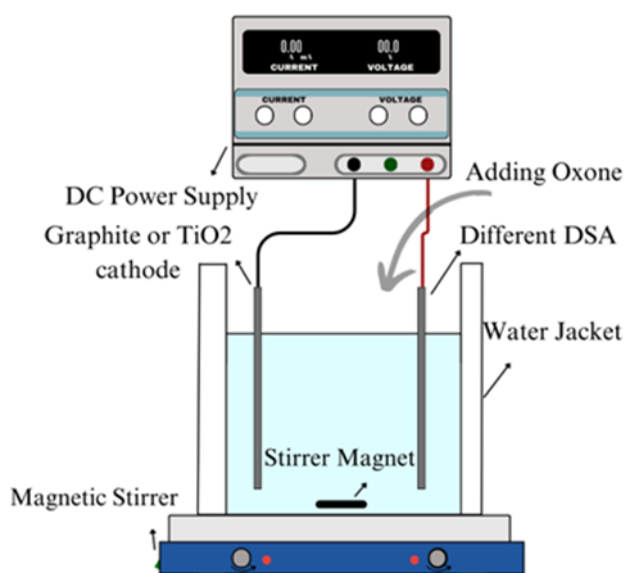


Figure 1. Schematic diagrams of EP post-treatment of EC pre-treated paper mill effluent

Experimental design and statistical analysis

RSM is a statistical and mathematical tool used to develop predictive models, design experiments, evaluate the influence of key factors and identify optimal operating conditions. In this study, RSM was used to optimise the EP process after EC pretreatment. A four-factor, three-level experimental design was used to investigate the influence of key operating parameters, including applied current (A), initial pH (B), PMS concentration (C) and reaction time (D). In addition, the response variables considered in the study were COD removal efficiency (Y_1) and specific energy consumption (SEC) (Y_2). Table 2 presents both the actual and coded values of the operating parameters used in the experimental design.

Table 2. Experimental design matrix and factor levels for the EO process combined with PMS

Symbol and variables	Coded factors		
	-1	0	1
A: Applied current (A)	1	3	5
B: Initial pH	2	6	10
C: PMS dosage (mM)	5	15	25
D: Time (min)	10	50	90

The experimental data were analyzed with the quadratic equation given in Eq. 6.

$$Y = \beta_0 + \sum \beta_i X_i + \sum \beta_{ii} X_i^2 + \sum_i \sum_j \beta_{ij} X_i X_j \quad (6)$$

In Eq. (6), Y represents the response variable of the design, β_0 denotes the intercept or constant term, while β_i , β_{ii} , and β_{ij} correspond to the linear, quadratic, and interaction coefficients, respectively. X_i and X_j are the independent operating parameters. COD removal efficiencies were calculated using the Eq. (7).

$$\text{Removal efficiency, \%} = \frac{C_0 - C_t}{C_0} \quad (7)$$

where C_0 and C_t are the initial and final COD concentrations after a reaction time of t.

Specific Energy Consumption

In order to assess the cost-effectiveness of the process, SEC values were determined under the optimum operating conditions identified in the study. Energy consumption was calculated using Eq. 8.

$$\text{SEC} = \frac{U \times I \times t}{V} \quad (8)$$

where, U is the cell voltage (V), I is the applied current (A) and t is the electrolysis time (h), while V is the volume of treated wastewater in cubic metres (m^3).

RESULTS AND DISCUSSION

Experimental Design and Model Development of COD Removal

The BBD results showed a maximum COD removal efficiency of 98.9% for EC pre-treated paper mill effluent. Notably, moderate values of SEC (481.5 kWh/m³) was also observed at the experimental condition corresponding to run 11, where this highest removal efficiency was achieved. On the other hand, the lowest COD removal efficiency (23.1%) was obtained in run 26 with the lowest SEC (13 kWh/m³) (Table 3).

Analysis of variance (ANOVA) was performed to evaluate the relationships between the independent variables and the response outcomes (Table 4). The coefficient of determination (R^2) was used to assess the fit of the polynomial model to the data, and the Fisher F-test was used to determine the statistical significance of the model. As seen in Table 4, the R^2 and adjusted R^2 (Adj- R^2) values for COD removal efficiency were 0.9615 and 0.9230, respectively. The "Predicted R^2 " value of 0.7956 is in good agreement with the "Adj R^2 " value of 0.9230, which indicates that the model has a strong predictive capability. With an F-value of 24.98, the model produces a statistically significant result, suggesting that it is highly reliable. There is only a 0.01% chance that such a high F-value would occur by chance or due to experimental noise. Model terms with "Prob > F" values less than 0.0500 are considered statistically significant. The "Lack of Fit F-value" of 3.33 indicates that the lack of fit is not statistically significant when compared to the pure error. This means there is a 12.90% probability that such a value could arise due to random variation (noise). A non-significant lack of fit is desirable, as it suggests that the model adequately fits the experimental data well. The significance of a model can be evaluated using the adequate precision and coefficient of variation (CV%) values. The "Adequate Precision" value, which reflects the signal-to-noise ratio, is 19.363. As a ratio above 4 is considered acceptable, this high value confirms that the model has sufficient signal strength and can be reliably used to explore and optimise the design space. While acceptable thresholds for C.V.% may vary depending on the field of study, in general, values below 10% indicate excellent model reliability. Values between 20% and 30% are considered acceptable, while values exceeding 30% suggest poor model precision and are typically deemed unacceptable [30, 31]. In this study, the C.V.% for COD removal efficiency was calculated as 9.55, indicating a high level of model reliability.

The results presented in Table 4 also show the p-values of each independent variable (A, B, C and D), as well as their quadratic (A^2 , B^2 , C^2 and D^2) and two-factor (AB, AC, AD, BC, BD and CD) terms. In this analysis, terms A and D were found to be highly significant, as were A^2 , C^2 and D^2 . Conversely, 'Prob > F' values greater than 0.1000 indicate non-significant term. The other independent variables (B and C), all of the two-factor variables, and the quadratic term of B^2 were found to be insignificant. The equation derived from the ANOVA results for COD removal efficiency, expressed in terms of the coded factors, is given by Eq. (9).

$$\begin{aligned} \text{COD (\%)} \text{ for coded factors} = & +82.38 + 19.34 A - 0.18 B + \\ & 1.74 C + 24.72 D + 0.20 AB + 0.17 AC + 2.27 AD + \\ & 0.67 BC - 2.03 BD + 0.13 CD - 12.96 A^2 - 2.93 B^2 - \\ & 10.13 C^2 - 12.93 D^2 \end{aligned} \quad (9)$$

Figure 2 shows the experimental and predicted values of COD removal efficiencies for EP process, indicating the adequacy of the normal probability model. The discrepancies between the predicted and actual results indicated that the quadratic regression model, which displayed a linear distribution pattern, was acceptable.

The contour plots and response surface graphs for COD removals are presented in Figure 3. As can be seen from the contour plots (Figures 3a, 3c and 3e), the COD removal efficiency increased as the applied current rose from 1 A to 3 A. However, increasing the current beyond 3 A did not significantly improve COD removal efficiency. Conversely, no improvement in COD removal efficiency was observed as the initial pH increased, suggesting that the initial pH did not affect COD removal in the EP process (Figure 3a and b). Similarly, increasing the PMS dosage and initial pH did not improve COD removal efficiency (Figure 3c and d). It can be said that the midpoint PMS dosage (15 mM) slightly improved the COD removal efficiency. As previously indicated, the process time was also found to be a highly significant parameter according to the quadratic model, with an increase in COD removal efficiency observed as the time increased from 10 to 90 min (Figures 3e and f).

In EO processes, especially those coupled with PMS activation, the efficiency of COD removal is heavily dependent on operational parameters. The applied current is critical because it governs the generation rate of reactive oxidants, such as $\bullet\text{OH}$ and $\text{SO}_4\bullet^-$ radicals, at the electrode surface. While increasing the current density generally promotes higher COD removal by enhancing the formation of these species, excessively high current levels may trigger competing side reactions, such as oxygen evolution. This lowers the overall current efficiency [32]. The pH of the solution strongly influences the speciation and stability of the generated oxidants. Under acidic conditions, sulfate radicals ($\text{SO}_4\bullet^-$) are preferentially formed, whereas hydroxyl radicals ($\bullet\text{OH}$) dominate in neutral to alkaline media. Therefore, the most effective COD removal is generally achieved within slightly acidic to neutral pH ranges depending on the predominant oxidation pathway [20, 33, 34]. Prolonging the reaction time usually improves COD removal because it allows more time for the complete oxidation of organic contaminants. However, beyond a certain point, improvement tends to plateau, due to factors such as depleted oxidants or mass transfer constraints [35]. During electrochemical activation, PMS acts as a precursor to generate sulfate radicals. Increasing the PMS dosage increases radical availability, promoting more effective COD degradation. However, excessive dosing can lead to radical quenching and increased chemical consumption, which diminishes the process's overall efficiency.

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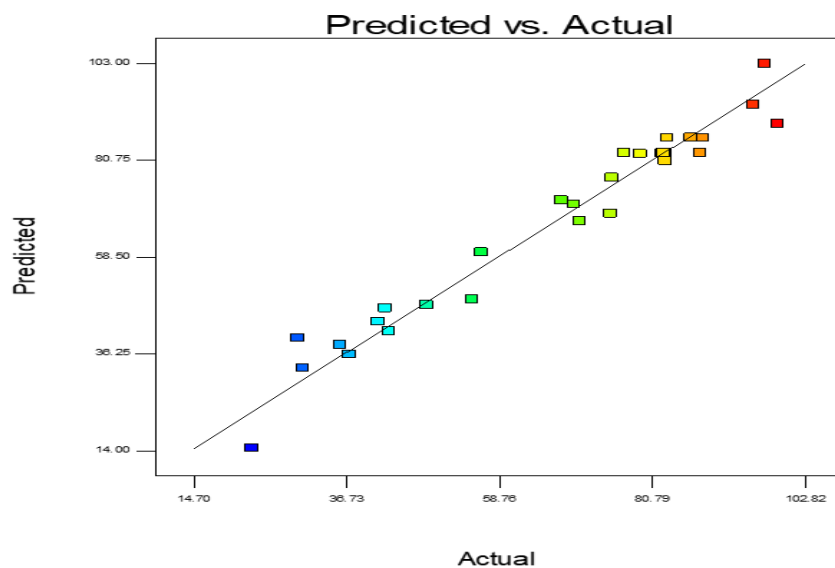


Figure 2. The experimental and predicted values of COD removal for EO combined with PMS

Table 3. Experimental conditions and responses (COD removal, SEC and SECCOD values)

Run	A: Applied current (A)	B: Initial pH	C: PMS (mM)	D: Time (min)	Y1: COD removal (%)	Voltage (V)	Y2: SEC value (kWh/m ³)
1	5	6	25	50	82.7	12.8	533.3
2	5	6	15	10	54.9	12.6	105.0
3	5	6	15	90	97.0	19.7	1477.5
4	3	2	15	90	95.4	10.7	481.5
5	3	10	5	50	70.3	11.9	297.5
6	3	6	15	50	82.7	10.0	250.0
7	3	2	5	50	74.8	11.6	290.0
8	5	2	15	50	88.1	13.8	575.0
9	3	6	15	50	76.8	10.9	272.5
10	3	2	25	50	69.5	9.6	240.0
11	3	10	15	90	98.9	10.7	481.5
12	5	10	15	50	82.9	14.3	595.8
13	1	6	5	50	35.8	7.5	62.5
14	1	10	15	50	42.3	0.7	5.8
15	3	6	25	90	86.4	10.4	468.0
16	3	6	15	50	87.7	11.9	297.5
17	3	6	5	90	79.1	12.6	567.0
18	1	6	15	90	56.2	7.4	111.0
19	3	10	25	50	67.8	9.7	242.5
20	3	6	5	10	30.4	13.4	67.0
21	1	6	25	50	42.8	7.3	60.8
22	1	2	15	50	48.3	6.8	56.7
23	5	6	5	50	75.0	14.2	591.7
24	3	10	15	10	41.3	12.0	60.0
25	3	6	15	50	82.5	11.2	280.0
26	1	6	15	10	23.1	7.8	13.0
27	3	6	15	50	82.2	9.9	247.5
28	3	6	25	10	37.1	10.9	54.5
29	3	2	15	10	29.7	11.9	59.5

Table 4. ANOVA for Response Surface Quadratic Model for COD removal

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F
Model	14003.65	14	1000.26	24.98	< 0.0001
A-Applied current	4490.49	1	4490.49	112.13	< 0.0001
B-Initial pH	0.40	1	0.40		0.9219
C-PMS	36.20	1	36.20	0,90	0.3579
D-Time	7331.29	1	7331.29	183,07	< 0.0001
AB	0.17	1	0.17		0.9494
AC	0.11	1	0.11		0.9583
AD	20.56	1	20.56	0,51	0.4854
BC	1.82	1	1.82	0.045	0.8343
BD	16.44	1	16,44	0,41	0.5321
CD	0.071	1	0.071		0.9670
A ²	1089.41	1	1089.41	27,20	0.0001
B ²	55.84	1	55.84	1,39	0.2573
C ²	665.96	1	665.96	16.63	0.0011
D ²	1084.50	1	1084.50	27.08	0.0001
Residual	560.64	14	40.05		
Lack of Fit	500.45	10	50.05	3.33	0.1290
Pure Error	60.19	4	15.05		
Cor Total	14564.29	28			

Table 5. ANOVA for Response Surface Quadratic Model for SEC

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F
Model	2.408E+006	14	1.720E+005	23.02	< 0.0001
A-Applied current	1.061E+006	1	1.061E+006	142.02	< 0.0001
B-Initial pH	31.69	1	31.69	4.241E-003	0.9490
C-PMS	398.19	1	398.19	4.241E-003	0.3714
D-Time	8.681E+005	1	8.681E+005	116.17	< 0.0001
AB	1284.03	1	1284.03	0.17	0.6848
AC	802.78	1	802.78	0.11	0.7479
AD	4.061E+005	1	4.061E+005	54.35	< 0.0001
BC	6.25	1	6.25	8.364E-004	0.9773
BD	0.063	1	0.063	8.364E-006	0.9977
CD	1870.56	1	1870.56	0.25	0.6246
A ²	37850.35	1	37850.35	5.07	0.0410
B ²	3706.01	1	3706.01	0.50	0.4928
C ²	1059.06	1	1059.06	0.14	0.7122
D ²	13758.58	1	13758.58	1.84	0.1963
Residual	1.046E+005	14	7472.17		
Lack of Fit	1.028E+005	10	10284.29	23.27	0.0041
Pure Error	1767.50	4	441.88		
Cor Total	1.570E+005	28			

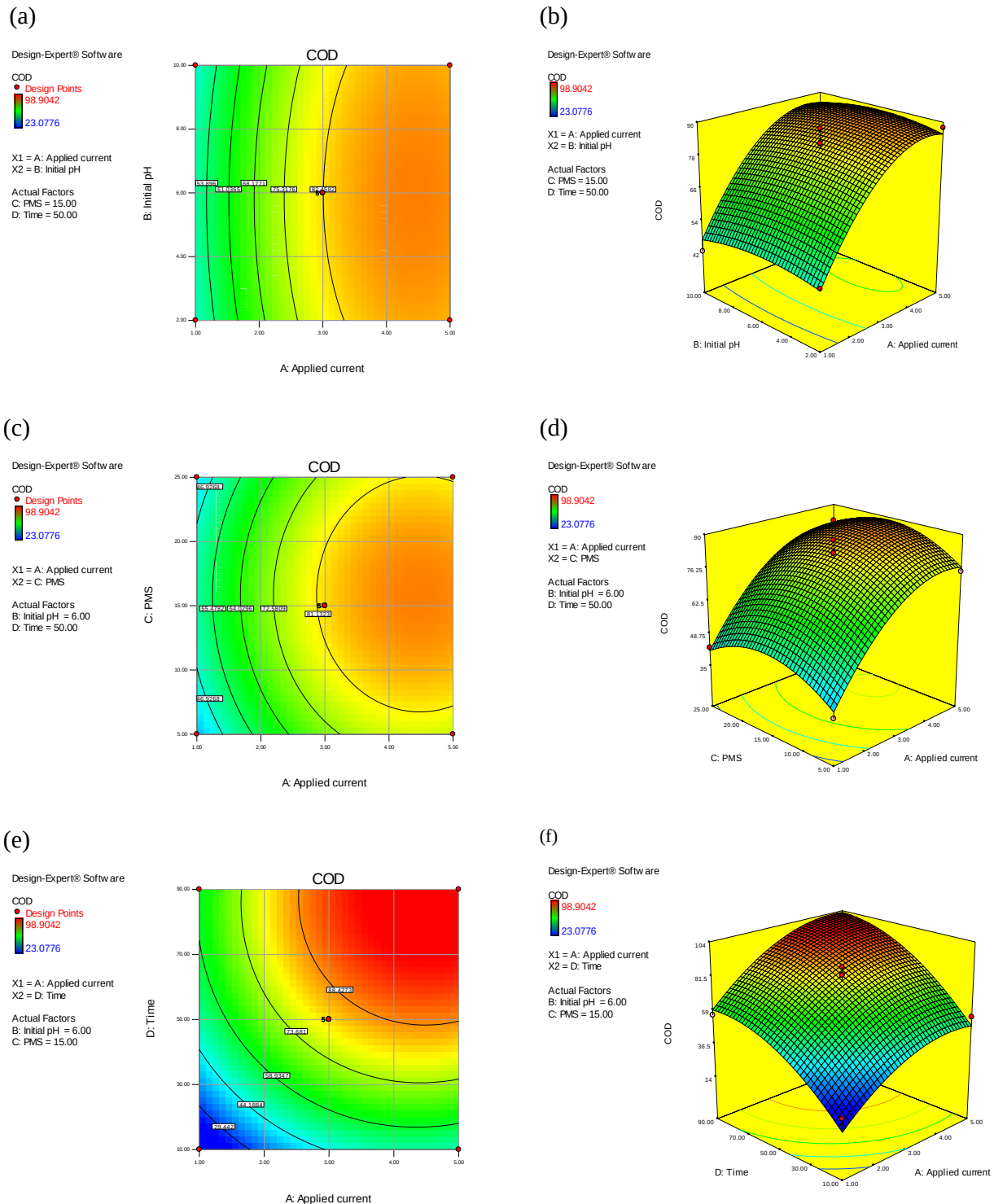


Figure 3. Response surface graphs showing the COD removal efficiency of the EP process

Experimental design and model development of SEC

Electrochemical treatment processes, such as EC and EO, are energy-intensive technologies that have a significant impact on overall energy consumption costs. The energy consumption of the EP post-treatment process was evaluated by calculating related SEC values for each batch tested in this study. The results of the ANOVA are presented in Table 5. The R^2 value of the quadratic model was found to be 0.9584. The Pred R^2 of 0.7632 is in reasonable agreement with the Adj R^2 of 0.9167, with both measures indicating a high degree of fit.

The model's F-value of 23.02 suggests that the model is statistically significant. It is thought that a "Model F-Value" this large could only occur due to noise in 0.01% of cases. Values of 'Prob > F' below 0.0001 indicate that the model terms are highly significant. "Adeq Precision" is evaluated by the signal-to-noise ratio, with values above 4 being considered desirable. With a ratio of 20.458, the signal is strong, confirming that the model is suitable for exploring and optimising the design space. The C.V. for the model was determined to be 28.34%, indicating a moderate level of variability in the data relative to the mean, which may suggest

potential improvement areas in model precision. With a "lack of fit F-value" of 23.27, there is only a 0.41% probability that such a high value could result from random variation (noise), indicating a statistically significant lack of fit. This suggests that the model does not accurately represent the data, which is undesirable as a good fit is essential for making reliable predictions.

The Prob > F values for the independent variable, its quadratic term and its two-factor term can be seen in Table 5. In this case A, D, AD, A² were found significant model terms. As is well known, values greater than 0.1000 suggest that the model terms are insignificant. Therefore, none of the other parameters significantly affect SEC values. The equation derived from the ANOVA results for SEC is given by Eq. (10).

$$SEC \left(\frac{kWh}{m^3} \right) \text{ for coded factors} = +269.50 + 297.38 A - 1.63 B - 23.04 C + 268.96 D + 17.92 AB - 14.17 AC + 318.63 AD - 1.25 BC - 0.13 BD - 21.63 CD + 76.39 A^2 - 23.90 B^2 - 12.78 C^2 + 46.06 D^2 \quad (10)$$

Figure 4 shows the experimental and predicted values of SEC values for the combined EO-PMS treatment process, demonstrating the normal probability model's suitability. The close agreement between the two sets of values suggests that the quadratic regression model, with its linear distribution trend, provides an acceptable fit to the data. Figure 5 shows contour and 3D response surface graphs for the SEC values in the EO-PMS process. As can be seen in Figure 5a-b and 5c-d, the applied current has a strong, direct influence on SEC values: energy consumption increases substantially as the current increases. In contrast, the effect of the initial pH or PMS concentration on the SEC is relatively

minor across the investigated range. Figure 5e-f shows how the applied current and reaction time interact to affect SEC. The surface plot shows that SEC increases sharply with increasing applied current and extended reaction time. The curvature of the surface indicates that these two parameters influence energy consumption synergistically, with the highest SEC values occurring at the maximum current and the longest reaction time. Therefore, minimising both current intensity and treatment duration is essential for achieving energy-efficient operation in the EP process.

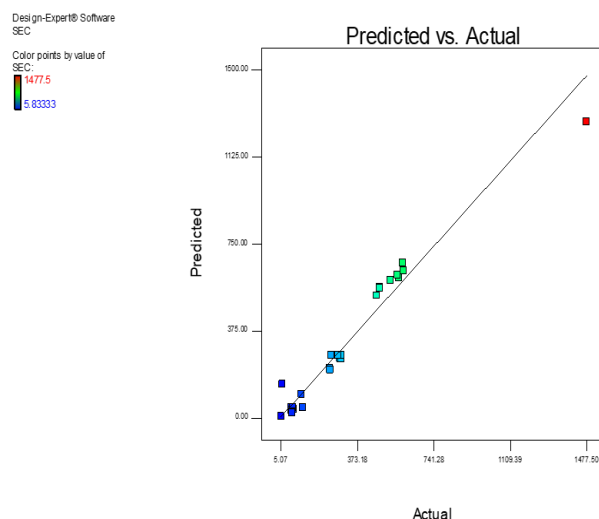


Figure 4. The experimental and predicted values of SEC values for EP process

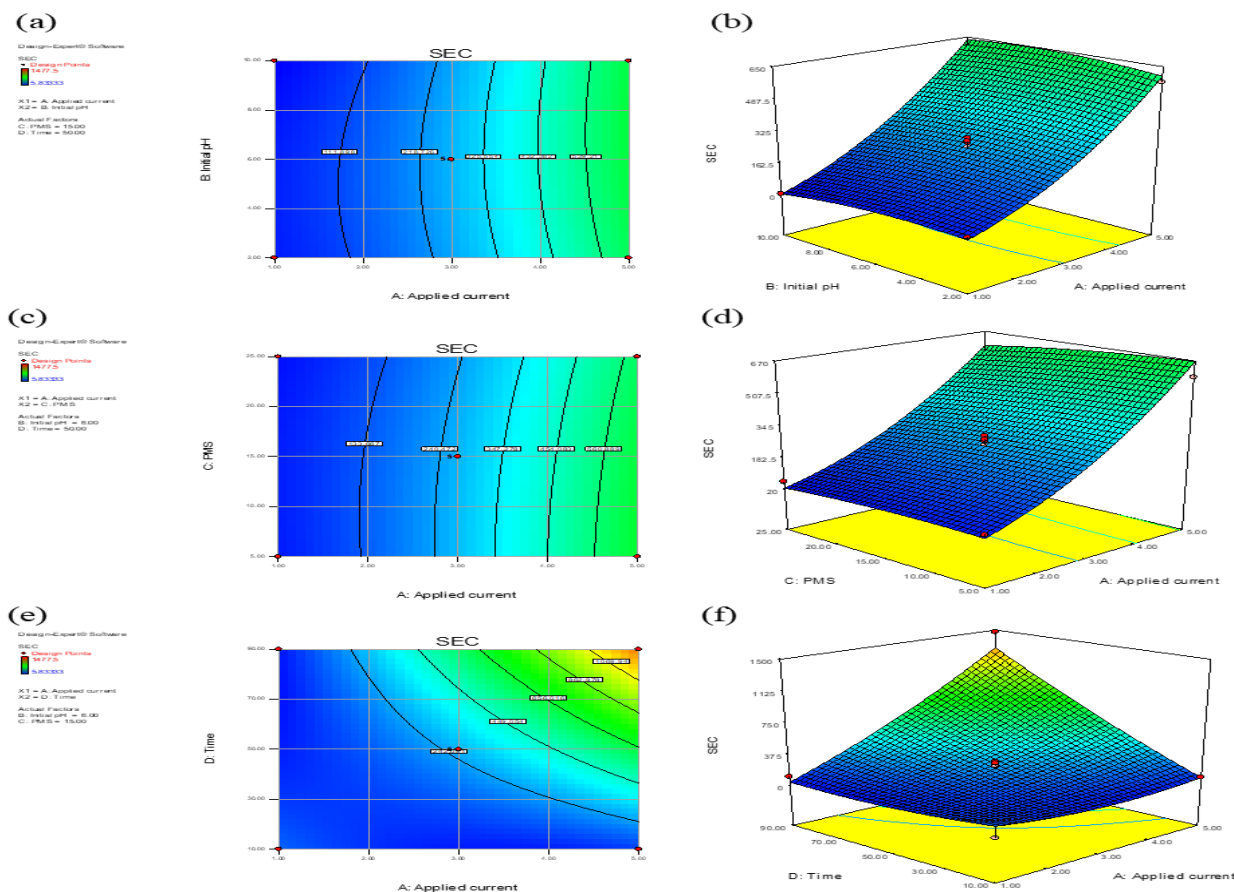


Figure 5. Response surface graphs showing the SEC values of the EO-PMS process

Determination of the Optimized Conditions and Verification Experiments

The optimization results of the EP process demonstrate how process parameters influence COD removal efficiency and SEC when treating paper industry wastewater. Setting the goal to maximize COD removal yielded optimal conditions of a current of 4.32 A, a pH of 8, a PMS dose of 16.3 mM and a reaction time of 80 minutes, with a predicted COD removal of 99.9% and a desirability value of 1.000 (Table 6). In practice, a COD removal efficiency of $92.3\% \pm 2.3$ was achieved, alongside $86.3\% \pm 2.8$ tannin/lignin removal and $57.2\% \pm 6.03$ additional colour removal in the post-treatment. While the experimental COD removal efficiency was slightly lower

than the model prediction, these results confirm the effectiveness of the optimised conditions, particularly with regard to degrading organic compounds. The overall COD removal efficiency was found to be 95.05% using hybrite EC and EO with PMS. On the other hand, the SEC values at pre- and post-treatment stages were calculated at 57.75 kWh/m³ and 790 kWh/m³, respectively. The higher SEC observed during the post-treatment stage can be attributed to persulfate activation and the use of Pt-based anodes, which are known to promote strong oxidative conditions. However, the substantial COD reduction (>95%) indicates that the increased energy input in the post-treatment stage effectively enhances the degradation of recalcitrant organic compounds, justifying the energy expenditure.

Table 6. Optimum operating conditions and results of verification experiment in the post-treatment

Optimum process parameter	Value	Desirability	Predicted	Experimental
Goal: Maximize COD				
Applied current (A)	4.32	1.000	COD: 99.9% removal	COD: $92.3\% \pm 2.3$
Initial pH	8			Tannin/Lignin: $86.3\% \pm 2.8$
PMS dose (mM)	16.3			Color: $57.2\% \pm 6.03$
Time (min)	80			

CONCLUSION

The findings of this study show that the EP process, when optimized through response surface methodology, is a very effective post-treatment strategy for pre-treated paper industry wastewater. The process achieved substantial removal of COD, tannin/lignin, and color. This was done under optimised conditions. This confirms its potential for the degradation of complex and recalcitrant organic pollutants. While a trade-off between treatment efficiency and energy consumption was observed, the optimization results suggest that the EP process can be adjusted according to treatment priorities, whether the goal is to maximize pollutant removal or minimize energy requirements. In addition, the statistical validation of the developed model shows that it is reliable for predicting and improving processes. These results show that EP post-treatment can be added to industrial wastewater treatment plans. This can help protect the environment more and meet the standards for discharging wastewater. Future studies could focus on how to use this technology on a larger scale and the costs and benefits of doing so to help companies use it.

DATA AVAILABILITY STATEMENT

The authors confirm that the data that supports the findings of this study are available within the article. Raw data that support the finding of this study are available from the corresponding author, upon reasonable request.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest with any individual, institution, or organization in the preparation, evaluation, or publication of this study.

USE OF AI FOR WRITING ASSISTANCE

Grammar and proof editing were refined using QuillBot and Grammarly in accordance with ethical writing standards.

ETHICS

There are no ethical issues with the publication of this manuscript.

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